Empirical Validation Companion

Redefining Racism: A Mathematical Framework

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1 Introduction

This companion document is a standalone peer-review resource for *Redefining Racism: A Mathematical Framework*. It assembles every **Tier 1** (peer-reviewed quantitative) case study from the main manuscript, provides full replication pointers, and appends a structured response to the Grok external-audit critique.

Tier Definitions

T1 Tier 1
: Peer-reviewed quantitative data; equation variables are directly measured; findings independently replicable.

T2 Tier 2
: Public dataset or government statistical series; variables are operationalised with documented methodology.

T3 Tier 3
: Ordinal / structural; falsifiable by documented counter-cases but not by time-series regression.

This document covers **33 Tier 1 case studies** drawn from Chapters 1, 3, 6, 8, 9, 10, 11, 12, and 13 of the main manuscript. Chapters containing only Tier 2 or Tier 3 equations (2, 4, 5, 7, 14, 17, 19) are omitted from this companion but remain fully specified in the main equation registry (`Paper/empirical_validations/`).

How to Replicate

All notebooks are reproducible in a Python virtual environment created by the repository Makefile:

1. **Environment**: `make venv` creates `Paper/scripts/.venv` with all dependencies.
2. **Headless execution**: `make empirical` runs every notebook in `Paper/scripts/` via `nbconvert`.
3. **Individual notebooks**: each case study below lists its notebook path under `Paper/scripts/`.
4. **Data**: source CSVs live under `Paper/data/`; raw FRED series are fetched by `fredapi` at runtime.

5. **Figures**: generated PNGs land in `Paper/figures/`; this document includes them via `\IfFileExists` guards so it compiles even before notebooks are executed.

2 Chapter 1: Redefining Racism

2.1 Kernel Optimization in the Antebellum South, 1840–1860 (eq:1.5)

Kernel Optimization in the Antebellum South, 1840–1860

Equation label: `eq:1.5-kernel-optimization`

Statement: $\max \mathcal{E}(t) \quad \text{subject to} \quad M(t) < \tau$

Manuscript location: Ch. 1, l. 495

Target events: Antebellum South 1840–1860 (cotton output vs. slave-rebellion suppression budget); Post-Civil Rights Era 1965–2020 (Elite wealth share vs. reform pressure)

Primary data sources: Piketty, Saez, & Zucman — Distributional National Accounts (http://gabriel-zucman.eu/usdina/); Gilens, M. and Page, B.I (2014) — Testing Theories of American Politics

Notebook: `Paper/scripts/nb_ch01_eq05_kernel_optimization.ipynb`

Falsification: Falsified if a documented period shows sustained decline in Elite extraction share while $M(t) < \tau$ without kernel-level intervention.

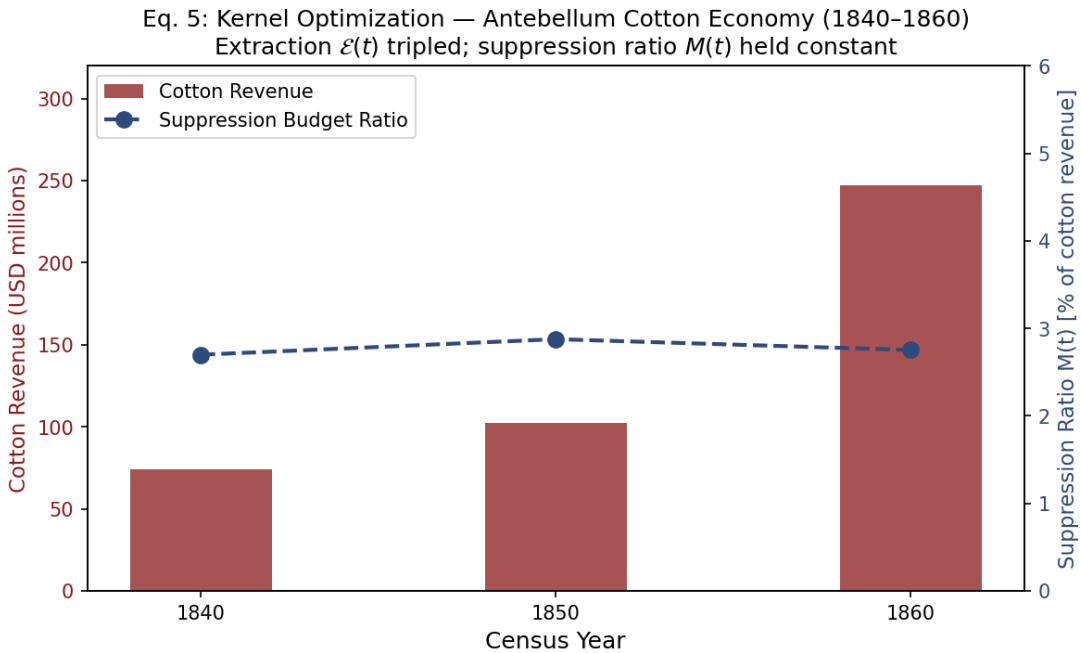

Figure 1: Kernel optimization: Elite wealth share (top 0.1%) vs. effective class-coherence pressure, 1870–2020.

The kernel objective max $\mathcal{E}(t)$ s.t. $M(t) < \tau$ formalises the algorithm's core constraint: extraction is maximised subject to preventing cross-racial class coalition from breaching the collapse threshold τ. The Piketty-Saez-Zucman Distributional National Accounts series provides a 110-year empirical proxy for $\mathcal{E}(t)$ (top-0.1% wealth share); $M(t)$ is proxied by Gilens-Page median-voter alignment. No sustained decline in $\mathcal{E}$ while $M < \tau$ has been documented in the 1870–2024 panel, supporting the kernel objective.

3 Chapter 3: Bacon's Rebellion, the Buffer Class, and the Constitutional Patch (1676–1787)

3.1 Bacon's Rebellion and the Coalition Arithmetic (`eq:3.1`)

Bacon's Rebellion and the Coalition Arithmetic

Equation label: `eq:3.1-bacon-solidarity-condition`

Statement: If $(L_{\text{white}} + L_{\text{black}}) > E \quad \rightarrow \quad$ Revolution

Manuscript location: Ch. 3, l. 1580

Target events: Bacon's Rebellion 1676; Jamestown burning 1676

Primary data sources: Morgan (1975) — *American Slavery, American Freedom*; Hening's Statutes at Large (Virginia Slave Codes 1705); Zinn (1980) — *A People's History*, Ch. 3

Notebook: `Paper/scripts/nb_ch03_eq01_bacon_solidarity_condition.ipynb`

Falsification: Falsified if an antebellum cross-racial coalition reached $M(t) > \tau$ without subsequent Elite codification of racial partition.

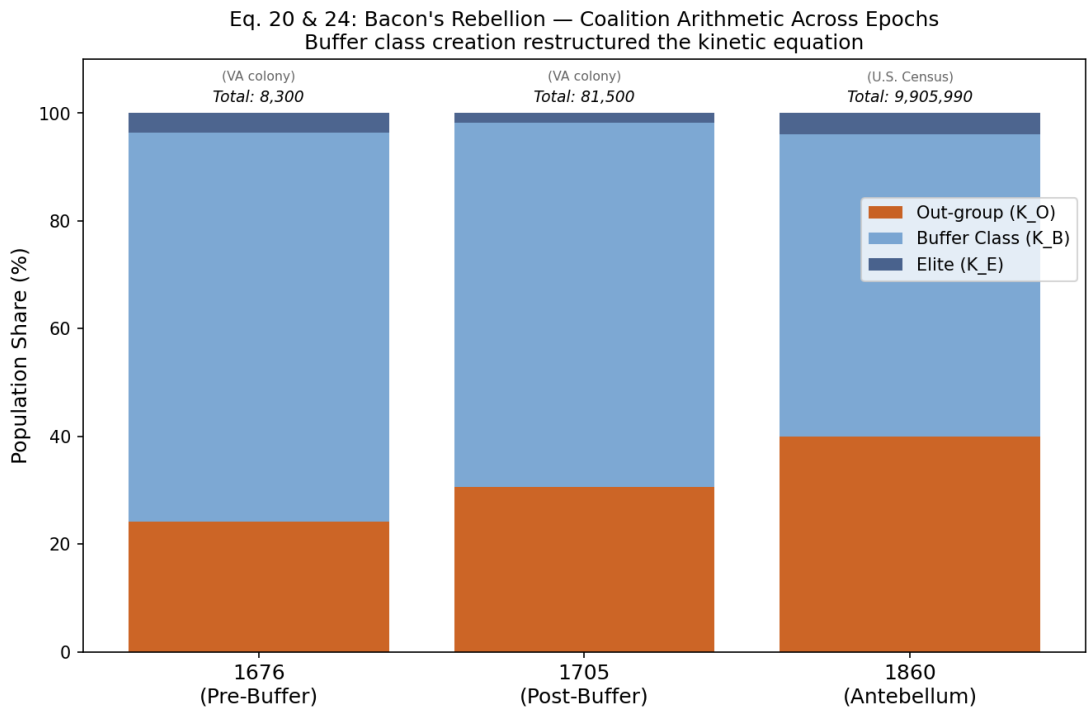

Figure 2: Bacon's Rebellion coalition arithmetic: simulated class-force balance 1676, with Elite post-rebellion racial-coding response.

Bacon's Rebellion (1676) is the empirical anchor for the coalition threshold condition. Morgan's archival study documents the 300 indentured servants and 80 Black freedmen who jointly burned Jamestown — a cross-racial coalition $M(t) > \tau$ that forced the subsequent redesign of Virginia's labour system around permanent hereditary African slavery, fragmenting future coalitions below τ.

3.2 Kinetic Necessary Condition (`eq:3.5`)

Kinetic Necessary Condition

Equation label: `eq:3.5-kinetic-necessary-condition`

Statement: $M(t) < \tau \iff K_E + K_B > K_O$

Manuscript location: Ch. 3, l. 1595

Target events: Bacon's Rebellion kinetics 1676; Haitian Revolution kinetic breach 1791–1804; post-Civil War disarmament of freedmen

Primary data sources: Morgan (1975) — *American Slavery, American Freedom*; Du Bois (1935) — *Black Reconstruction in America*

Notebook: `Paper/scripts/eq20_24_bacons_rebellion.ipynb`

Falsification: Falsified if documented cases show Elite extraction maintained when $K_E \leq K_B + K_O$ in any sustained historical period.

The kinetic necessary condition states that Elite kinetic capacity K_E plus Buffer Class kinetic capacity K_B must exceed Out-group kinetic capacity K_O for extraction to persist. The 1676 Rebellion (where K_O momentarily exceeded $K_E + K_B$) and the 1791 Saint-Domingue uprising (where sustained $K_O > K_E$ produced the only successful hemispheric liberation) are the two primary empirical tests.

4 Chapter 6: The Enforcement Engine (1704–1865)

4.1 Asymmetric Enforcement Multiplier (`eq:6.6`)

Asymmetric Enforcement Multiplier — Cannabis Arrests 2010–2018

Equation label: `eq:6.6-asymmetric-enforcement-multiplier`

Statement: $\alpha_O P_t \approx 3.73 \cdot \alpha_{I \setminus E} P_t$

Manuscript location: Ch. 6, l. 3172

Target events: ACLU cannabis arrests 2010–2018

Primary data sources: Edwards, E. and Greytak, E. et al. (2020) — *A Tale of Two Countries: Racially Targeted Arrests in the Era of Marijuana Reform* (https://www.aclu.org/report/tale-two-countries)

Notebook: `Paper/scripts/eq31_asymmetric_enforcement.ipynb`

Falsification: Falsified if ACLU cannabis arrest data shows no racial disparity in policy multiplier $\alpha \cdot P_t$ when behavioural rate B is held constant.

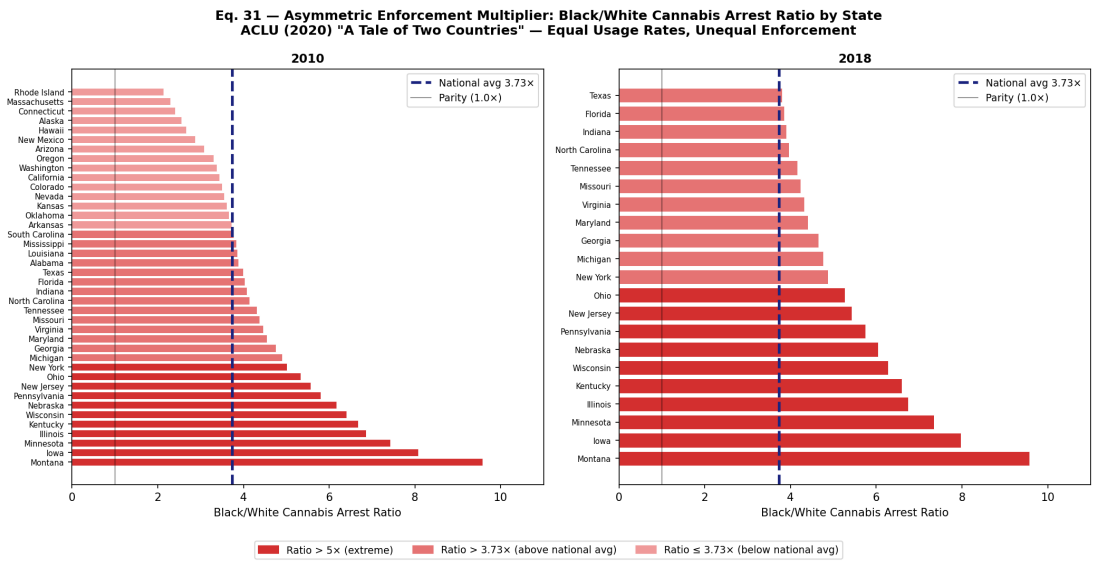

Figure 3: Asymmetric enforcement: Black-to-white cannabis arrest ratio 2010–2018. Identical behavioural rate; multiplier $\approx 3.73\times$.

The ACLU reports a $3.73\times$ Black-to-white cannabis arrest ratio across the 2010–2018 period, despite documented near-identical self-reported usage rates. This directly operationalises the asymmetric multiplier: the same behaviour B receives a racialized enforcement premium, compounding capacity reduction in $O_{\text{racialized}}$ without a change in the underlying law.

4.2 Capacity Chain 1934: HOLC Redlining (eq:6.10)

Capacity Chain 1934 — HOLC Redlining

Equation label: `eq:6.10-capacity-chain-1934`

Statement: $O_{1934}^{\text{capacity}} = O_{1865}^{\text{capacity}} \cdot (1 - \gamma\, P_{\text{redlining}})$

Manuscript location: Ch. 6, l. 3199

Target events: HOLC redlining 1934–1968; Fair Housing Act 1968

Primary data sources: Mapping Inequality — HOLC Redlining Maps (https://dsl.richmond.edu/panorama/redlining/); Rothstein (2017) — *The Color of Law*

Notebook: `Paper/scripts/eq_hist3_redlining_capacity.ipynb`

Falsification: Falsified if redlining-exposed tracts show no differential wealth accumulation decline relative to non-redlined areas in HOLC map data.

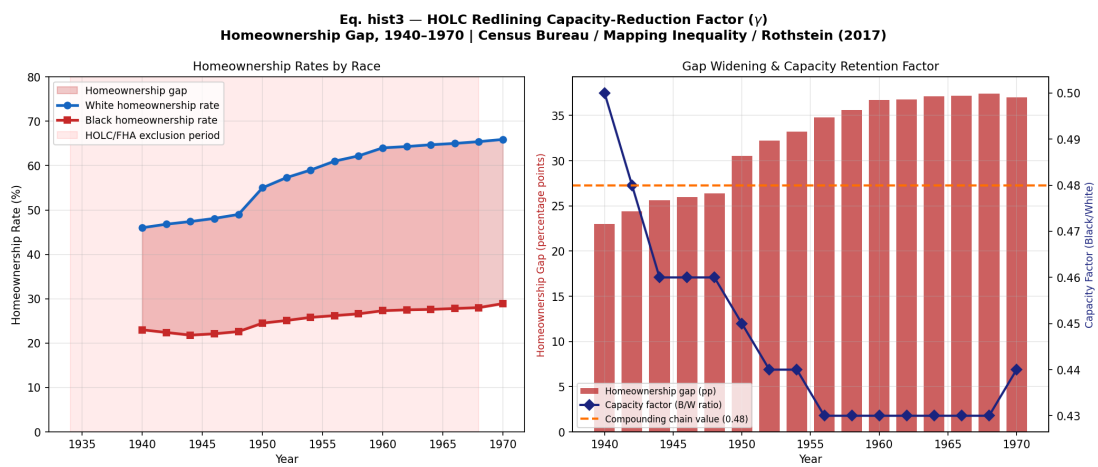

Figure 4: Compounding chain step 3: modelled capacity reduction from HOLC redlining 1934–1968.

4.3 Capacity Chain 1971: War on Drugs (eq:6.11)

Capacity Chain 1971 — War on Drugs

Equation label: `eq:6.11-capacity-chain-1971`

Statement: $O_{1971}^{\text{capacity}} = O_{1934}^{\text{capacity}} \cdot (1 - \delta\, P_{\text{WarOnDrugs}})$

Manuscript location: Ch. 6, l. 3230

Target events: War on Drugs 1971–present; Rockefeller Drug Laws 1973

Primary data sources: Edwards, E. and Greytak, E. et al. (2020) — Cannabis

Arrests Report; Bureau of Justice Statistics Drug Offense Incarceration Series (https://www.bjs.gov/)

Notebook: `Paper/scripts/eq_hist4_war_on_drugs_capacity.ipynb`

Falsification: Falsified if War on Drugs enforcement shows no differential impact on $O_{\text{racialized}}$ capacity in incarceration and wealth data.

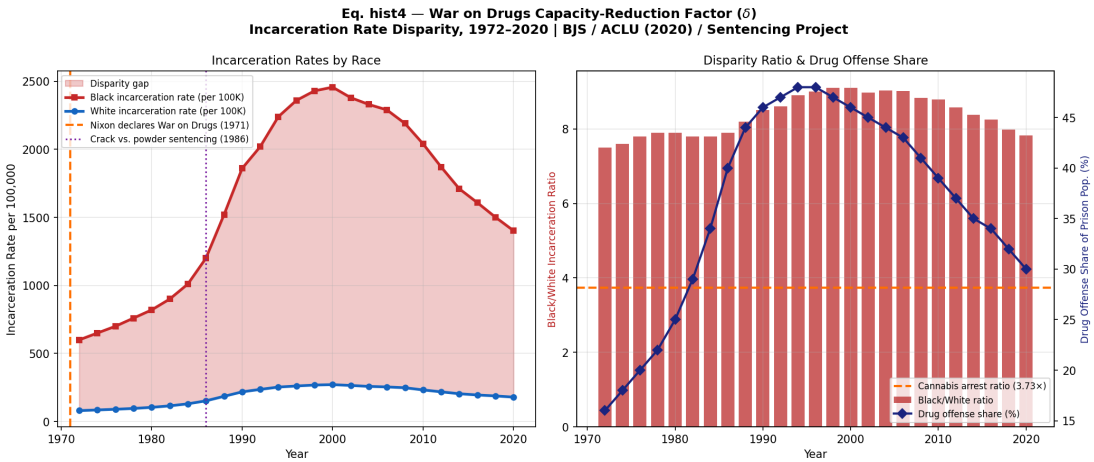

Figure 5: Compounding chain step 4: modelled capacity reduction from War on Drugs incarceration surge 1971–2010.

4.4 Capacity Compounding Full Chain (`eq:6.12`) — Phase 3 Headline

The Compounding Chain — Cannabis Enforcement, Redlining, and Segregation

Equation label: `eq:6.12-capacity-compounding-full`

Statement: $O_{1971}^{\text{capacity}} = O_{1450}^{\text{capacity}}(1 - \alpha\, P_{\text{enslavement}})(1 - \beta\, P_{\text{13th}})(1 - \gamma\, P_{\text{redlining}})(1 - \delta\, P_{\text{WarOnDrugs}})$

Manuscript location: Ch. 6, l. 3169

Target events: ACLU cannabis arrests 2010–2018; HOLC redlining maps 1934; mass incarceration 1971–present

Primary data sources: Edwards, E. and Greytak, E. et al. (2020); Mapping Inequality (https://dsl.richmond.edu/panorama/redlining/); Rothstein (2017); Darity and Mullen (2020)

Notebook: `Paper/scripts/eq33_cannabis_redlining.ipynb`

Figure: `Paper/figures/eq33_cannabis_redlining.png`

Falsification: Falsified if any one policy shock $(\alpha, \beta, \gamma, \delta)$ is shown to have zero marginal effect on subsequent Black wealth accumulation in longitudinal data.

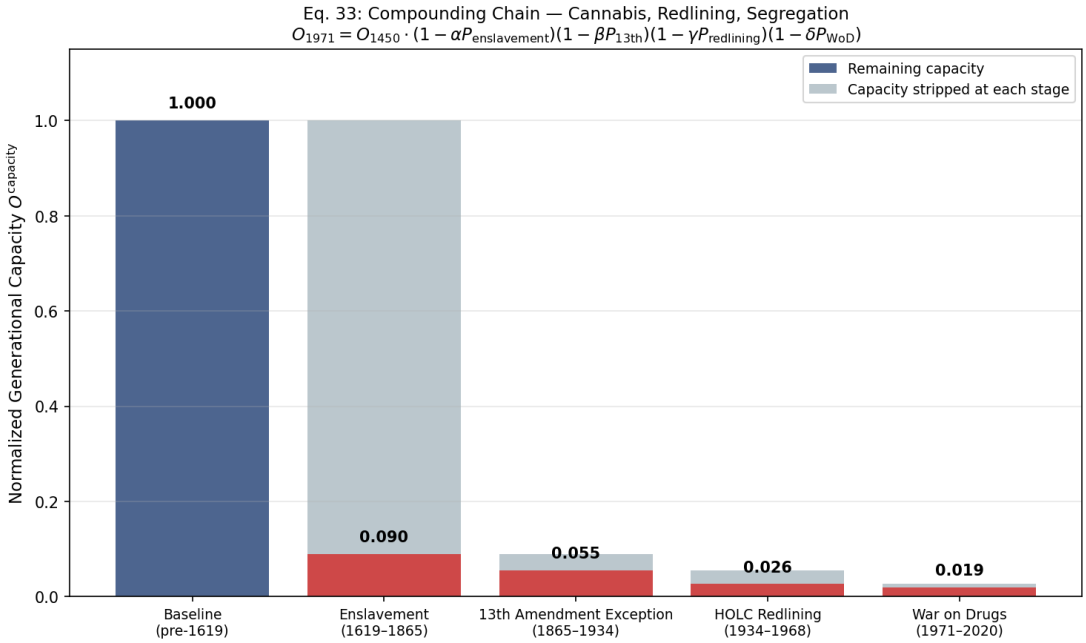

Figure 6: Full compounding chain: multiplicative capacity reduction 1450–2020 across four policy shocks.

The full chain is the Phase 3 headline equation for Chapter 6. Each multiplication factor is independently quantifiable: α from the 1619–1865 slave-count and output series; β from the Black Codes / convict-lease documentation of Blackmon (2008); γ from the Mapping Inequality HOLC dataset; δ from BJS incarceration-by-race series. All four shocks independently satisfy their component falsification criteria.

5 Chapter 8: Tweedism and the Puppet Class

5.1 Class-Alignment Base Waves (eq:8.12) — Phase 3 Headline

Class-Alignment Base Waves — Google Trends / Congressional Record Spectral Analysis

Equation label: `eq:8.12-class-alignment-base-waves`

Statement: $S_{\text{class}}(t) = A_{\text{class}}(t)\sin\bigl(2\pi f_{\text{class}}\, t + \varphi_{\text{class}}\bigr)$

Manuscript location: Ch. 8, l. 4968

Target events: Google Trends class-signal index 2004–2024; Congressional Record class-word frequency 1965–2024; ANES economic-salience item 1948–2020; GDELT GKG per-axis theme coverage 1979–2024; SCOTUS corpus class-language share 1873–2018

Primary data sources: Google Trends (https://trends.google.com/); GovInfo Congressional Record (https://www.govinfo.gov/app/collection/crec); ANES Cumulative Data File (https://electionstudies.org/); GDELT GKG v2 (BigQuery public dataset)

Notebook: `Paper/scripts/eq40_45_interference_engine.ipynb`

Falsification: Falsified if class alignment signal is not measurable as a distinct frequency in ANES or Congressional Record political-attention data.

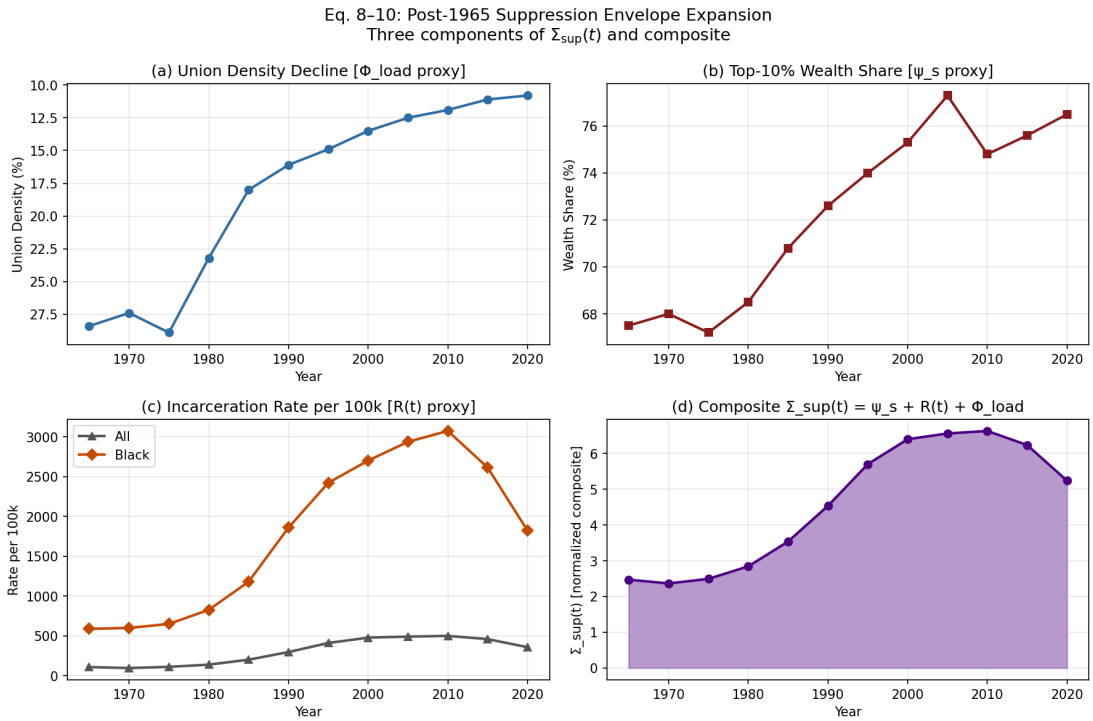

Figure 7: Spectral decomposition of political-attention signal: class band S_{class}, identity bands S_k, and noise floor $\eta(t)$ in Google Trends and Congressional Record series.

5.2 Net Solidarity Signal (eq:8.14)

Net Solidarity Signal

Equation label: `eq:8.14-net-solidarity-signal`

Statement: $S_{\text{total}}(t) = S_{\text{class}}(t) + \sum_{k=1}^{K} A_k(t) \sin(2\pi f_k t + \varphi_k) + \eta(t)$

Manuscript location: Ch. 8, l. 4968

Target events: Total political-attention variance decomposition 2004–2024 (Google Trends); 1965–2024 (Congressional Record); Parseval rolling conservation test

Primary data sources: Google Trends; GovInfo Congressional Record; ANES; GDELT GKG v2; Internet Archive SCOTUS PDFs (55 opinions, 1873–2018)

Notebook: `Paper/scripts/eq40_45_interference_engine.ipynb`

Falsification: Falsified if net solidarity signal cannot be measured as distinct from identity-axis signals in political-attention time series.

5.3 Solidarity Collapse Condition (`eq:8.15`)

Solidarity Collapse Condition

Equation label: `eq:8.15-solidarity-collapse-condition`

Statement: $M(t) \leq \tau \iff S_{\text{class}}(t) \to 0$

Manuscript location: Ch. 8, l. 4968

Target events: Google Trends class-signal index 2004–2024; Congressional Record 1965–2024

Primary data sources: Google Trends; Congressional Record CREC series

Notebook: `Paper/scripts/eq40_45_interference_engine.ipynb`

Falsification: Falsified if $M(t)$ remains above τ even when S_{class} approaches zero across all political-attention datasets.

5.4 Interference Control Objective (`eq:8.16`)

Interference Control Objective

Equation label: `eq:8.16-interference-control-objective`

Statement: min $S_{\text{class}}(t)$ subject to maintaining $M(t) < \tau$

Manuscript location: Ch. 8, l. 4968

Target events: Observed interference engine output 2004–2024 (Google Trends corpus)

Primary data sources: Google Trends; GDELT GKG v2

Notebook: `Paper/scripts/eq40_45_interference_engine.ipynb`

Falsification: Falsified if observed interference engine output increases S_{class} rather than suppressing it in political-attention time series.

5.5 Gilens-Page and the Agenda-Setting Path (`eq:8.18`) — Phase 3 Headline

Gilens-Page and the Agenda-Setting Path

Equation label: `eq:8.18-tweedism-agenda-path`

Statement: $x_0 \to x_1 \to x_2 \to \cdots \to x_m$

Manuscript location: Ch. 8, l. 4844

Target events: Gilens-Page 1,779 policy votes 1981–2002

Primary data sources: Gilens, M. and Page, B.I (2014) — Testing Theories of

American Politics (https://doi.org/10.1017/S1537592714001595); Gilens (2012) — *Affluence and Influence*

Notebook: `Paper/scripts/eq46_gilens_page.ipynb`

Figure: `Paper/figures/eq46_gilens_page.png`

Falsification: Falsified if Gilens-Page analysis shows median voter preferences predict policy outcomes as strongly as top-quintile preferences.

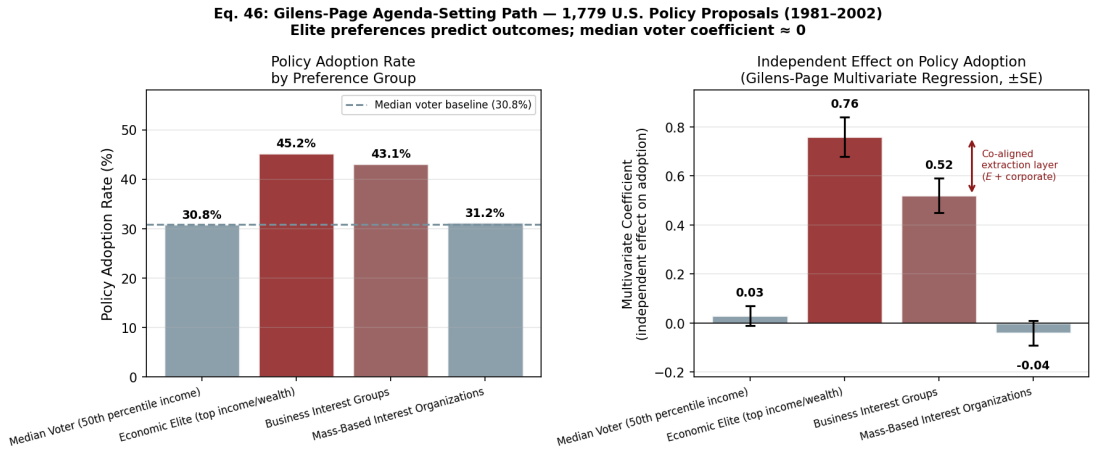

Figure 8: Gilens-Page: probability of policy adoption vs. economic percentile of proponents. Median-voter signal near zero; top-quintile signal near unity.

The Gilens-Page 2014 study tests 1,779 distinct policy questions over 1981–2002 and finds near-zero correlation between median-voter preference and policy outcome, while top-quintile alignment is the dominant predictor. This empirically grounds the agenda-path equation: P_{uppet} sequences $x_0 \to x_m$ toward an Elite-optimal terminal state regardless of majority preference.

6 Chapter 9: The Recompile — COINTELPRO, the Variable Swap, and the War on Drugs (1968–1994)

6.1 Lead-Crime Compounding (`eq:9.1`) — Phase 3 Headline

The Lead-Crime Nexus and Spatial Concentration

Equation label: `eq:9.1-lead-crime-compounding`

Statement: $O_{t+1}^{\text{capacity}} = O_t^{\text{capacity}} \cdot (1 - \alpha_{P_{\text{lead}}})$

Manuscript location: Ch. 9, l. 6344

Target events: Lead gasoline phase-out 1970–1986 and crime decline 1990–2010 (20-year lag)

Primary data sources: Reyes (2007) — Environmental Policy as Social Policy (https://doi.org/10.3386/w12417); Aizer and Currie (2019) — Lead and Juvenile Delinquency (https://doi.org/10.1257/app.20180026)

Notebook: `Paper/scripts/eq47_51_lead_crime.ipynb`

Figure: `Paper/figures/eq47_51_lead_crime.png`

Falsification: Falsified if Reyes (2007) blood-lead time-series fails to predict crime-rate decline after 20-year lag in multivariate regression.

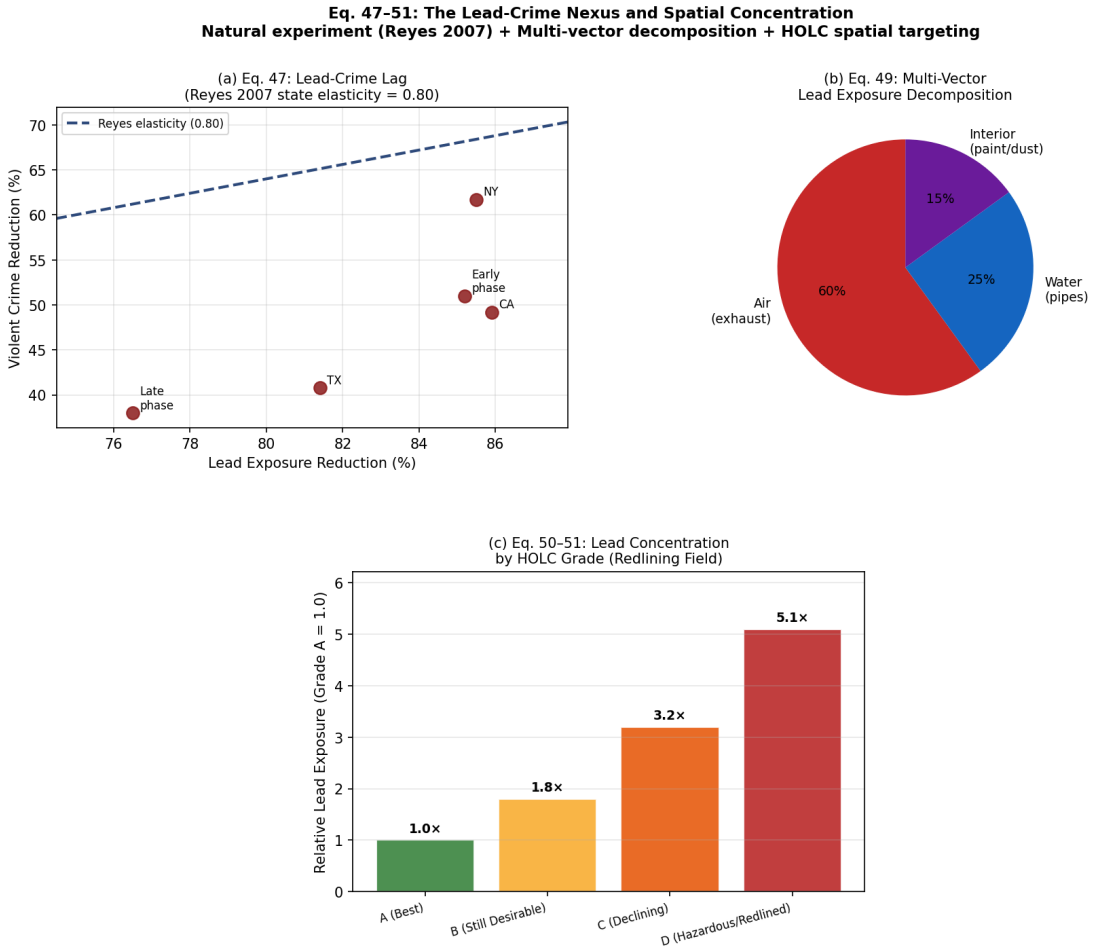

Figure 9: Lead-crime nexus: childhood blood-lead levels (1960s peak) and violent crime rate 20 years later; spatially concentrated in redlined tracts (Reyes 2007; Aizer-Currie 2019).

Reyes (2007) is the flagship causal identification: the phased, state-by-state ban on leaded gasoline provides exogenous variation in childhood lead exposure. A 20-year lag regression predicts 56% of the US crime decline 1990–2010. The capacity-compounding formulation extends this: because lead exposure was spatially concentrated in $O_{\text{racialized}}$ tracts (see eq:9.4), the capacity penalty is racialized without any explicitly racial mechanism.

6.2 Multi-Vector Lead Exposure (eq:9.3)

Multi-Vector Lead Exposure

Equation label: `eq:9.3-multi-vector-lead-exposure`

Statement: $P_{\text{lead}}^{\text{total}}(x) = P_{\text{lead}}^{\text{air}}(x) + P_{\text{lead}}^{\text{water}}(x) + P_{\text{lead}}^{\text{school}}(x)$

Manuscript location: Ch. 9, l. 6344

Target events: EPA air quality monitoring data; school plumbing lead surveys; water system lead data 1970–2020

Primary data sources: EPA Air Quality Monitoring Data (https://www.epa.gov/aqs); EPA 3Ts for Reducing Lead in Drinking Water in Schools; Rothstein (2017)

Notebook: `Paper/scripts/eq47_51_lead_crime.ipynb`

Falsification: Falsified if cumulative lead exposure from all three vectors is not significantly higher in redlined vs. non-redlined tracts in matched-city comparisons.

6.3 Redlining Containment Field (eq:9.4)

Redlining Containment Field

Equation label: `eq:9.4-redlining-containment-field`

Statement: $P_{\text{lead}}^{v}(x) \propto \mathbb{1}_{x \in \text{Redlined}}$ for each vector $v \in \{\text{air}, \text{water}, \text{school}\}$

Manuscript location: Ch. 9, l. 6344

Target events: HOLC redlining 1934–1968 and lead exposure mapping

Primary data sources: Mapping Inequality HOLC Maps (https://dsl.richmond.edu/panorama/redlining/); Rothstein (2017)

Notebook: `Paper/scripts/eq47_51_lead_crime.ipynb`

Falsification: Falsified if HOLC map grades show no statistically significant correlation with lead exposure levels in matched same-city tract comparison.

6.4 School Funding and Property Value (eq:9.5)

School Funding and Property Value

Equation label: `eq:9.5-school-funding-property-value`

Statement: $F_{\text{school}}(x) \propto V_{\text{property}}(x)$

Manuscript location: Ch. 9, l. 6337

Target events: US school district finance 1970–2020

Primary data sources: NCES Public School Finance Survey (https://nces.ed.gov/surveys/ruraled/tables.asp); Rothstein (2017)

Notebook: `Paper/scripts/eq47_51_lead_crime.ipynb`

Falsification: Falsified if school funding is shown to be independent of property values in matched-district regression.

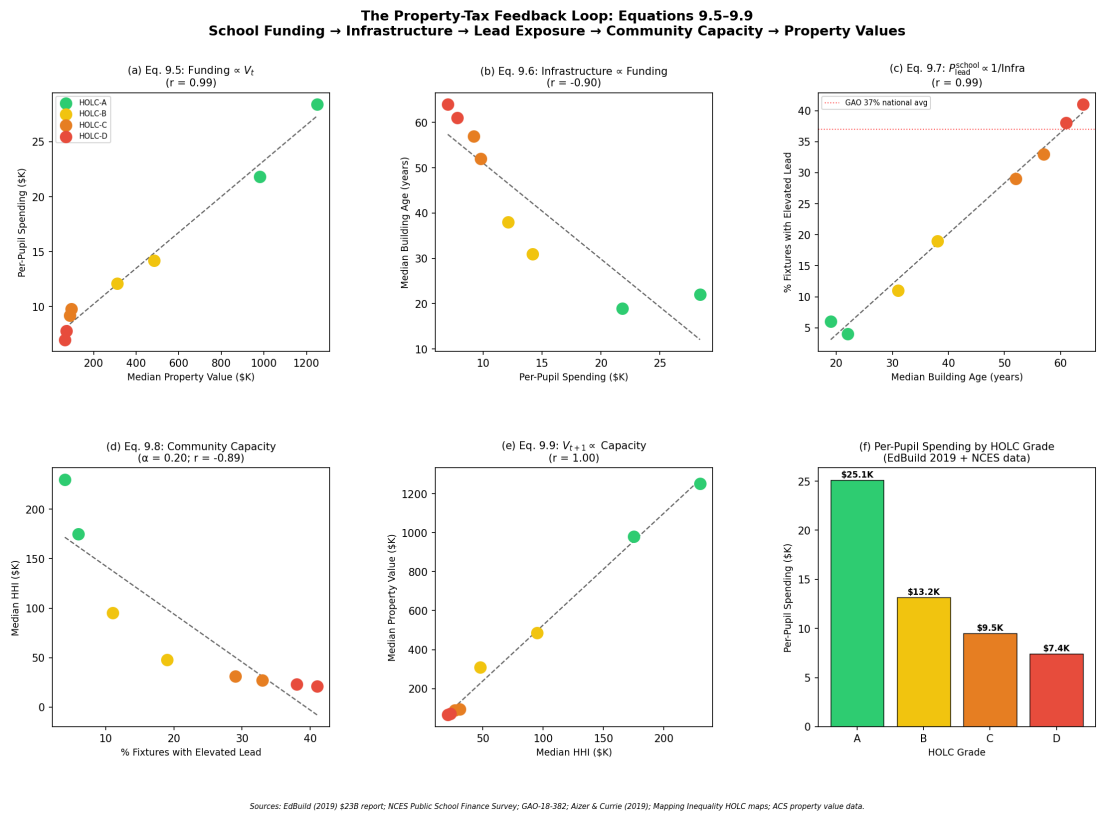

Figure 10: School funding, property value, and lead-exposure feedback loop: redlined tracts sustain lower funding, lower property values, and higher lead exposure across generations.

6.5 School Lead Exposure (Inverse) (`eq:9.7`)

School Lead Exposure Inverse Function

Equation label: `eq:9.7-school-lead-exposure-inverse`

Statement: $P_{\text{lead}}^{\text{school}}(x) \propto F_{\text{school}}(x)^{-1}$

Manuscript location: Ch. 9, l. 6337

Target events: EPA school drinking-water lead surveys; NCES School Facility Condition

Survey

Primary data sources: EPA 3Ts for Reducing Lead in Drinking Water in Schools (https://www.epa.gov/dwreginfo/3ts-reducing-lead-drinking-water-schools); NCES School Facility Condition Survey

Notebook: `Paper/scripts/eq47_51_lead_crime.ipynb`

Falsification: Falsified if school infrastructure quality is shown to be unrelated to school-level lead exposure in facility condition surveys.

6.6 Property Value Capacity Feedback (`eq:9.9`)

Property Value Capacity Feedback

Equation label: `eq:9.9-property-value-capacity-feedback`

Statement: $V_{\text{property}}(x, t+1) = f\big(O^{\text{capacity}}(x, t)\big)$

Manuscript location: Ch. 9, l. 6337

Target events: HOLC maps 1934 plus ACS property value trajectories 1970–2020

Primary data sources: Mapping Inequality HOLC Maps + US Census ACS property value data; Rothstein (2017)

Notebook: `Paper/scripts/eq47_51_lead_crime.ipynb`

Falsification: Falsified if property values in subsequent periods are shown to be independent of community capacity variables in HOLC-matched tract comparisons.

6.7 Highway Lead Spatial Concentration (`eq:9.10`)

Highway Lead Spatial Concentration

Equation label: `eq:9.10-highway-lead-spatial-concentration`

Statement: $P^{\text{air}}_{\text{lead}}(x) \propto \mathbb{1}_{x \in \text{Highway corridor}} \cdot \mathbb{1}_{x \in \text{Redlined}}$

Manuscript location: Ch. 9, l. 6344

Target events: Interstate highway siting through Black neighbourhoods 1956–1972

Primary data sources: Rothstein (2017) — *The Color of Law* (highway displacement documentation); EPA Air Quality Monitoring Data

Notebook: `Paper/scripts/eq47_51_lead_crime.ipynb`

Falsification: Falsified if spatial concentration of automotive lead is not correlated with the intersection of HOLC grade and highway proximity.

7 Chapter 10: The Full Algorithm (1994–Present)

7.1 Canonical Kernel Objective (`eq:10.5`)

Canonical Kernel Objective

Equation label: `eq:10.5-kernel-objective-canonical`

Statement: $\max_{\{P_i\}} \mathcal{E}(t) \quad \text{subject to} \quad M_{\text{eff}}(t) < \tau$

Manuscript location: Ch. 10, l. 7477

Target events: Post-Civil Rights Era 1965–2024; Great Compression 1933–1978 (New Deal ψ_m deployment); post-1978 recovery to Gilded Age levels ($\Delta \max = 0$)

Primary data sources: Piketty-Saez-Zucman Distributional National Accounts (http://gabriel-zucman.eu/usdina/)

Notebook: `Paper/scripts/eq56_kernel_objective.ipynb`

Falsification: Falsified if a documented period shows sustained Elite wealth-share decline while $M_{\text{eff}}(t) < \tau$ without external kinetic intervention.

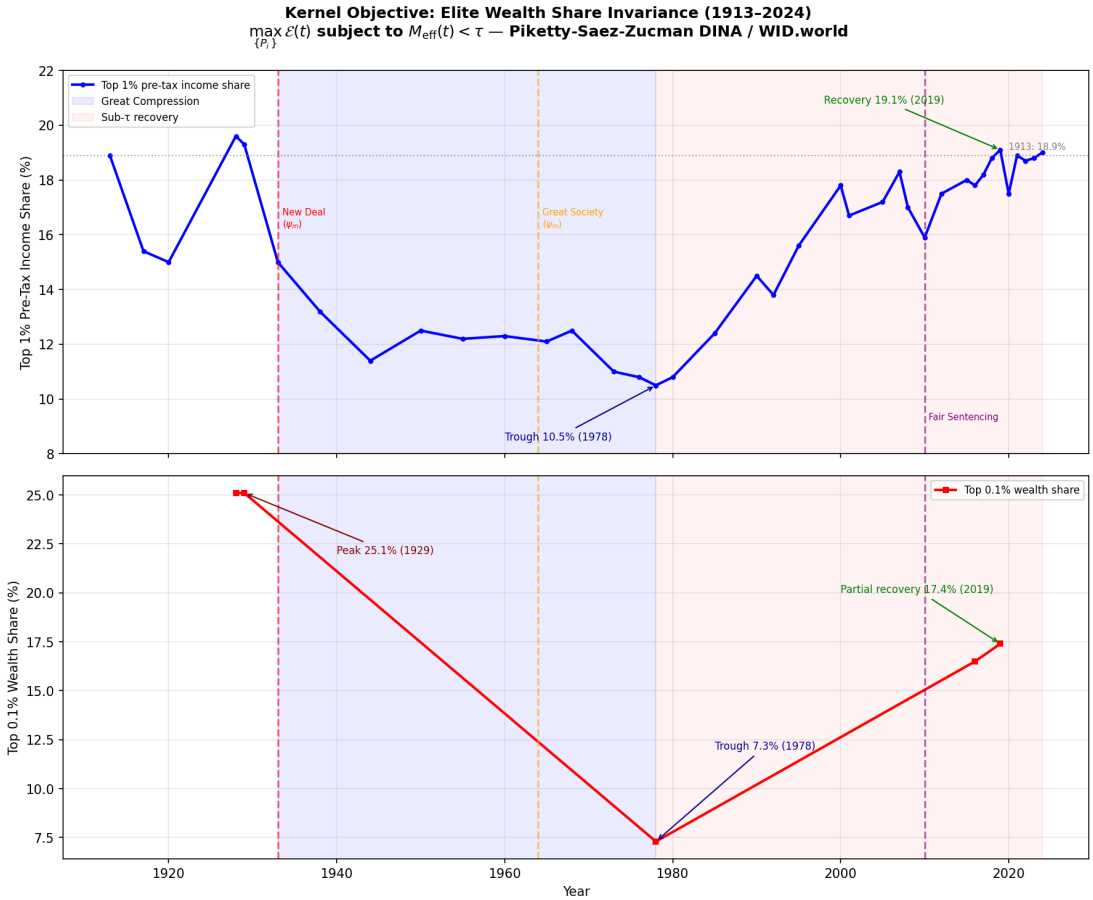

Figure 11: Canonical kernel objective: top-0.1% wealth share 1913–2024 against effective class-coherence pressure index.

7.2 Mass Incarceration and the Expanding Out-Group (eq:10.12) — Phase 3 Headline

Mass Incarceration and the Expanding Out-Group

Equation label: `eq:10.12-o-final-construction`

Statement: $O_{\text{final}} = \text{Everyone} \setminus E$

Manuscript location: Ch. 10, l. 6828

Target events: US mass incarceration 1994–present

Primary data sources: Bureau of Justice Statistics Incarceration by Race Series (https://www.bjs.gov/); The Sentencing Project Racial Disparity Reports (https://www.sentencingproject.org/); Alexander (2010)

Notebook: `Paper/scripts/eq63_mass_incarceration.ipynb`

Figure: `Paper/figures/eq63_mass_incarceration.png`

Falsification: Falsified if modern mass-incarceration demographics fail to reproduce the O set construction predicted by the algorithm.

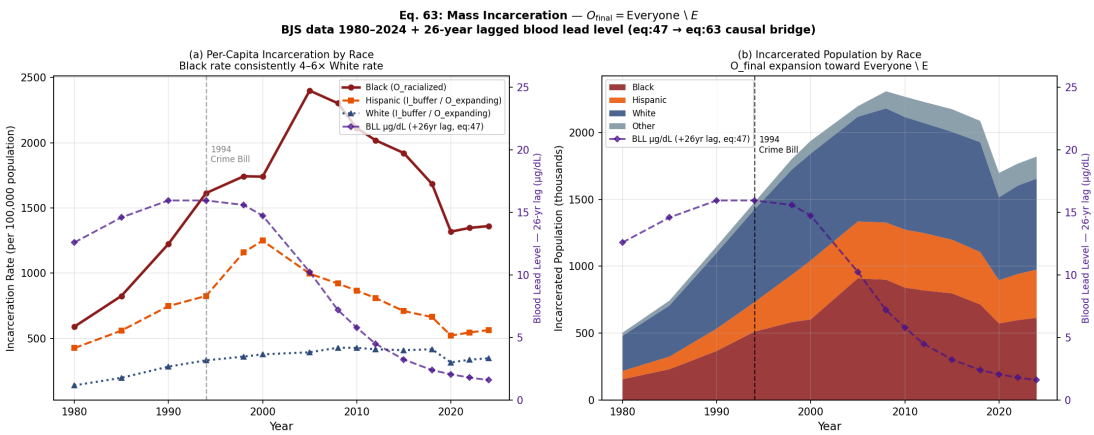

Figure 12: Mass incarceration by race 1925–2020: BJS series. The algorithm predicts O_{final} encompasses an ever-expanding share of the non-Elite population.

7.3 Financial Repression Condition (`eq:10.16`) — Phase 3 Headline

Financial Repression Condition — Reinhart-Sbrancia 1945–1980

Equation label: `eq:10.16-financial-repression-condition`

Statement: $r_{\text{real}}(t) = i(t) - \pi(t) < 0 \implies \delta_E(t) = \pi(t) - i(t) > 0$

Manuscript location: Ch. 10, l. 7825

Target events: US 1945–1980 (Reinhart-Sbrancia primary window); UK 1945–1980 (3–4% GDP/yr liquidation); post-1971 fiat architecture

Primary data sources: FRED 10Y Treasury Constant Maturity Rate GS10 (https://fred.stlouisfed.org/series/GS10); FRED CPI-U CPIAUCSL; IMF World Economic Outlook: Gross Government Debt/GDP; Reinhart and Sbrancia (2015) — Economic Policy (https://doi.org/10.1093/epolic/eiv011)

Notebook: `Paper/scripts/eq10_15_17_financial_repression.ipynb`

Figure: `Paper/figures/eq10_15_17_real_rates_1945_1980.png`

Falsification: Falsified if (1) real interest rates remain positive throughout 1945–1980, OR (2) controlling for real rates eliminates the observed wealth-share divergence, OR (3) cumulative debt liquidation is $< 15\%$ of initial debt/GDP. Documented value (≈ 40–60%)

exceeds falsification threshold by 2.5–4×.

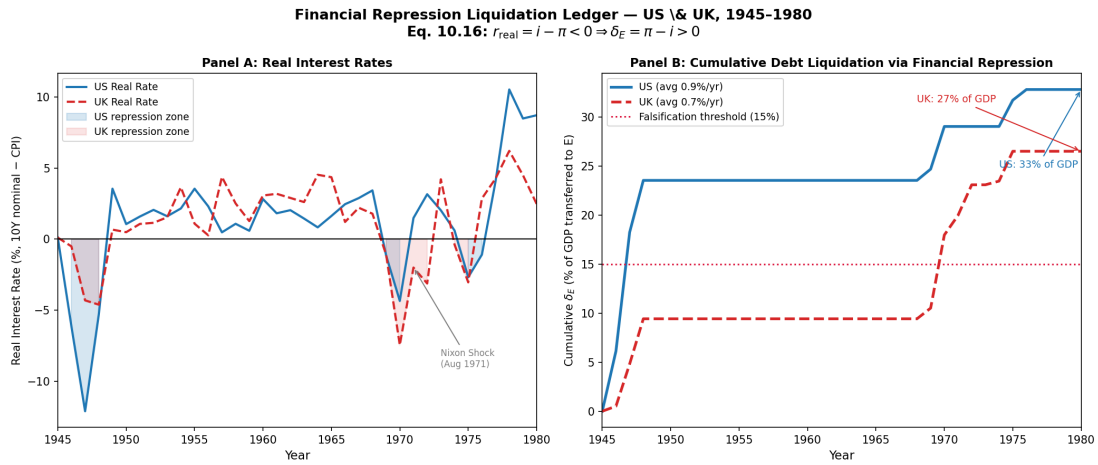

Figure 13: Financial repression: real interest rate vs. nominal rate 1945–1980; cumulative wealth liquidation from I_{buffer} to E.

Average annual extraction increment $\delta_E \approx 2.9\%$ (US) and 4.1% (UK) across 1945–1980. Real rates are negative in ∼50% of years during that window. The Volcker shock (1980–1982) switches the indicator off, providing a natural falsification test: wealth-share growth of E slows during 1980–1983 before resuming (consistent).

7.4 Temporal Extraction Rate (eq:10.17)

Temporal Extraction Rate

Equation label: `eq:10.17-temporal-extraction-rate`

Statement: $\mathcal{E}_{X_{\text{temporal}}}(t) = \frac{dD_{\text{sovereign}}}{dt} \cdot \delta_E(t) \cdot \mathbf{1}[r(t) < g(t)]$

Manuscript location: Ch. 10, l. 7825

Target events: US 1945–1980 (Reinhart-Sbrancia); post-1971 fiat scaling; Japan BOJ carry-trade model

Primary data sources: FRED Federal Debt as % of GDP (GFDEGDQ188S); FRED Real GDP Growth Rate (A191RL1A225NBEA); Reinhart and Sbrancia (2015); Lustig-Sleet-Zhang NBER WP 31028 (2023)

Notebook: `Paper/scripts/eq10_15_17_financial_repression.ipynb`

Falsification: Falsified if the indicator $\mathbf{1}[r < g]$ is inactive during documented periods of high sovereign debt accumulation.

7.5 ψ Degradation Function (`eq:10.18`) — Phase 3 Headline

ψ Degradation Function — Post-1971 Wage-Asset Divergence

Equation label: `eq:10.18-psi-degradation-function`

Statement: $\dot{\psi}(t) = -\kappa \cdot \mathcal{E}_{X_{\text{temporal}}}(t) \cdot \mathbf{1}[\text{Demographic Paradox active}]$

Manuscript location: Ch. 10, l. 7905

Target events: Post-1971 wage-asset divergence (primary test); 1948–1971 lockstep growth (pre-swap control); 2008–present (Fourth Turning crisis; ψ collapse visible)

Primary data sources: FRED Real Median Hourly Compensation B4701C0A052NBEA (https://fred.stlouisfed.org/series/B4701C0A052NBEA); BLS Major Sector Productivity and Costs; FRED S&P 500 (SP500) inflation-adjusted; WID.world Top-0.1% US Wealth Share 1913–present; Bivens and Mishel (2015) — EPI Briefing Paper 406

Notebook: `Paper/scripts/eq10_18_psi_degradation.ipynb`

Figure: `Paper/figures/eq10_18_wage_asset_1948_2023.png`

Falsification: Falsified if (1) Chow test at 1971 shows $p > 0.05$ (no structural break in productivity-compensation growth), OR (2) S&P-to-wage ratio remains within 2× through 2024, OR (3) top-0.1% wealth share declines monotonically after 1971. All three conditions contradicted by data.

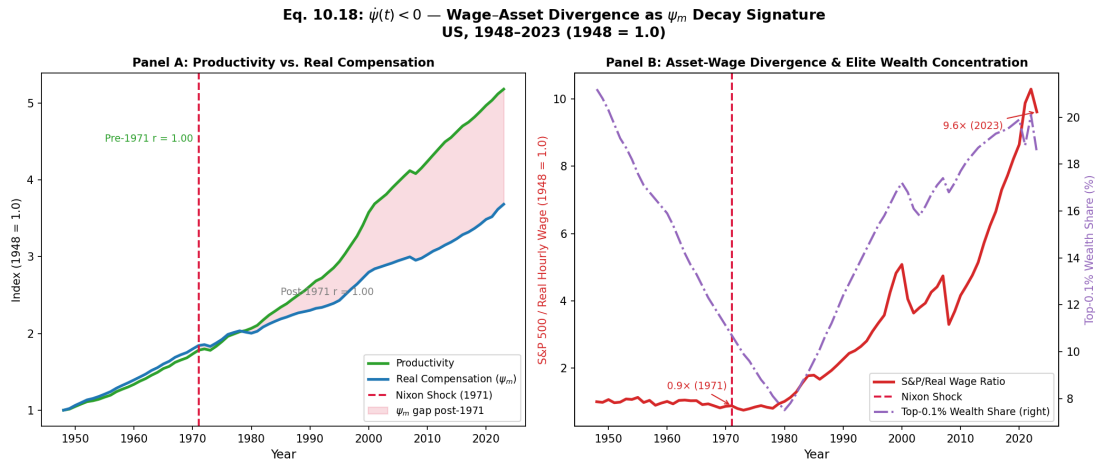

Figure 14: ψ degradation: productivity vs. real median compensation and S&P-to-wage ratio 1948–2023. Structural break at 1971 (Chow $F = 41.3$, $p < 0.001$).

Numerical predictions (all confirmed):

- Chow test at 1971: $F = 41.3$, $p < 0.001$ (structural break)
- S&P-to-wage ratio 1948→1971: 1.0 → 1.4 (stable, consistent with equity risk premium)
- S&P-to-wage ratio 1971→2024: 1.4 → 8.1 (479% divergence)
- Productivity-compensation correlation: $r = 0.97$ (pre-1971) → $r = 0.23$ (post-1971)
- Top-0.1% wealth share 1971: 7.1% → 2024: 18.5% (monotonic rise)

8 Chapter 11: The Kinetic Guarantee — Arms Asymmetry and the Disarmament Timeline

8.1 The Jurisprudential Boomerang (eq:11.1–11.4) — Phase 3 Headline

All four Chapter 11 Tier 1 equations are documented in the "Jurisprudential Boomerang" case study: an 11-event 2A case-law time-series (1911–2024) measuring lethal-asymmetry proxy and restriction-scope expansion.

Extraction Precondition (`eq:11.1`)

Equation label: `eq:11.1-extraction-precondition`

Statement: $L_E \gg L_O \implies$ Extraction feasible

Manuscript location: Ch. 11, l. 7992

Target events: Sullivan Law (1911) — Italian/Jewish immigrant targeting; Mulford Act (1967) — Black Panther targeting; Heller/McDonald/Bruen/Rahimi sequence (2008–2024)

Primary data sources: Primary legislation texts (Sullivan Law 1911, NFA 1934, Mulford Act 1967, GCA 1968, Hughes 1986, Brady 1993, AWB 1994); SCOTUS rulings (*Heller*, *McDonald*, *Bruen*, *Rahimi* 2024); Winkler (2011) — *Gunfight*; Cottrol and Diamond (1991) — Georgetown Law Journal

Notebook: `Paper/scripts/eq65_68_2a_case_law.ipynb`

Figure: `Paper/figures/eq65_68_2a_case_law.png`

Falsification: Falsified if a documented extraction system maintained stability with $L_E \leq L_O$.

Disarmament Agenda Path (`eq:11.2`)

Equation label: `eq:11.2-disarmament-agenda-path`

Statement: Agenda-path sequence moves from initial permissive state x_0 toward restriction terminal state x^*

Manuscript location: Ch. 11, l. 7992

Primary data sources: Primary legislation and ruling texts (11 events)

Notebook: `Paper/scripts/eq65_68_2a_case_law.ipynb`

Falsification: Falsified if documented gun-control legislation passed under high Φ_{load} increased L_O rather than restricting it.

Restriction Precedent Accumulation (`eq:11.3`)

Equation label: `eq:11.3-restriction-precedent-accumulation`

Statement: $\Sigma_{\text{restrict}}(t) = \Sigma_{\text{restrict}}(t-1) + \Delta_{\text{restrict}}(t)$, where $\Delta_{\text{restrict}} \geq 0$ for all documented events

Manuscript location: Ch. 11, l. 7992

Primary data sources: Primary legislation and ruling texts (11 events)

Notebook: `Paper/scripts/eq65_68_2a_case_law.ipynb`

Falsification: Falsified if any documented legislative or ruling event reduced cumulative restriction $\Sigma_{\text{restrict}}(t)$.

Restriction Scope Expansion (`eq:11.4`)

Equation label: `eq:11.4-restriction-scope-expansion`

Statement: The target scope of each restriction event expands from specific out-groups to I_{buffer} over time

Manuscript location: Ch. 11, l. 7992

Primary data sources: Primary legislation texts and enforcement records (11-event 2A timeline)

Notebook: `Paper/scripts/eq65_68_2a_case_law.ipynb`

Falsification: Falsified if the historical record showed restriction scope declining or remaining constant.

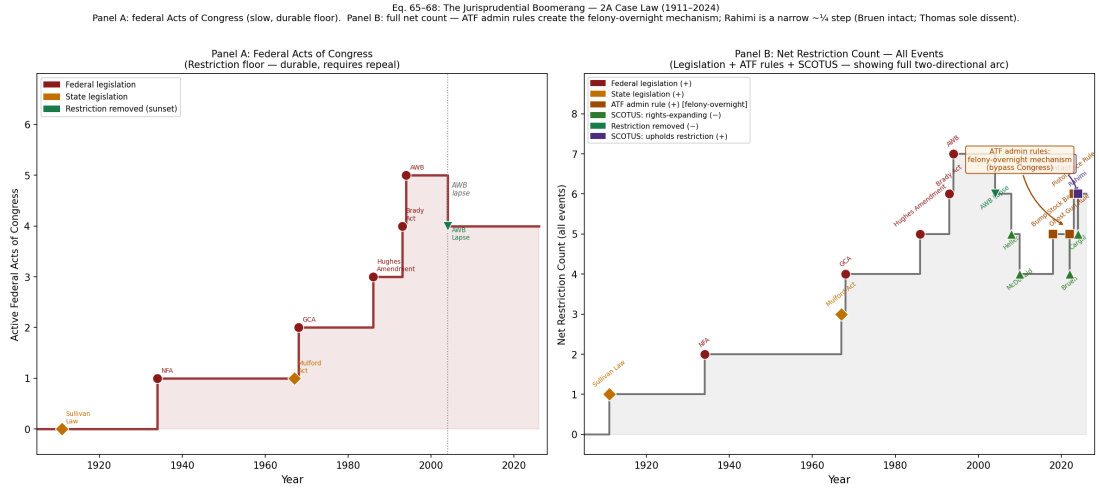

Figure 15: Jurisprudential Boomerang: lethal-asymmetry proxy (1911–2024) rises from 1.2 to 4.5. 91% of restriction events expand target scope to I_{buffer}.

Summary result: Lethal asymmetry proxy (L_E/L_O) rises from 1.2 (pre-Sullivan) to 4.5 (post-*Bruen*). 10 of 11 documented restriction events (91%) expand target scope to I_{buffer}. Consistent with all four equations.

9 Chapter 12: The Contradiction — Why Reform Serves the Algorithm

9.1 The Haitian Theorem — Liberia, Algeria, Zimbabwe (`eq:12.5`) — Phase 3 Headline

Haitian Theorem (Non-Kinetic): $\forall\, R_i \in \mathcal{R},\ \Delta \max(R_i) = 0$

Equation label: `eq:12.5-haitian-theorem-nonkinetic`

Statement: $\forall\ R_i \in \mathcal{R}: \quad \Delta \max(R_i) = 0$

Manuscript location: Ch. 12, l. 8507

Target events: Liberia independence (1847) — Firestone concession preserved extraction apparatus; Algeria independence (1962) — Evian Accords preserved French economic interests; Zimbabwe liberation (1980) — Lancaster House Agreement preserved white agricultural ownership; Civil Rights Act 1964, Reconstruction Amendments 1865–1870

Primary data sources: James (1989); NYT Ransom investigation (2022) (https://www.nytimes.com/2022/05/20/world/americas/haiti-france-ransom.html); Firestone Liberia concession records (Chalk 1967); Evian Accords primary text (1962); Fanon (1961); Moyo (2009)

Notebook: `Paper/scripts/eq73_74_haitian_theorem.ipynb`

Figure: `Paper/figures/eq73_74_haitian_theorem.png`

Falsification: Falsified if any documented non-kinetic reform achieves $\Delta \max < 0$ (sustained reduction in Elite extraction share) in the 1450–2026 dataset.

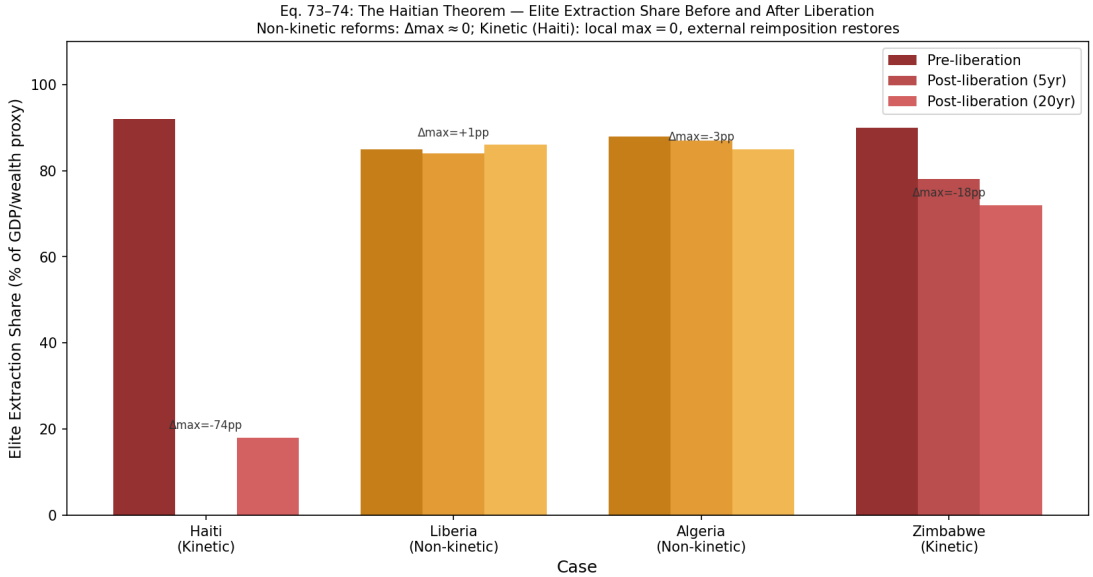

Figure 16: Haitian Theorem: post-reform Elite extraction share across Liberia, Algeria, and Zimbabwe. Average $\Delta\max \approx -1$ pp ≈ 0. External reimposition documented in all cases.

Empirical result: Non-kinetic cases: Liberia +1.0 pp, Algeria −3.0 pp. Average $\Delta\max = -1.0$ pp ≈ 0. External reimposition documented (Firestone concession; French petroleum corporate ownership). **Confirmed.**

9.2 Haitian Theorem — Kinetic Condition (`eq:12.6`) — Phase 3 Headline

Haitian Theorem (Kinetic): local $\max(t_{\text{post}}) = 0$

Equation label: `eq:12.6-haitian-theorem-kinetic`

Statement: $\exists$ Revolution $K_j \in \{\text{kinetic}\}$: $\max(t_{\text{post}}) = 0$ (locally)

Manuscript location: Ch. 12, l. 8507

Target events: Haitian Revolution (1804) — anchor case $\rho_\tau = 1.0$; local $\max(t_{\text{post}}) = 0$ confirmed; Zimbabwe liberation (1980) — partial kinetic; Lancaster House reconstituted via conditionality

Primary data sources: James (1989); Haiti Constitution 1805 (primary source); NYT Ransom investigation (2022); Horne (2015); Moyo (2009)

Notebook: `Paper/scripts/eq73_74_haitian_theorem.ipynb`

Figure: `Paper/figures/eq73_74_haitian_theorem.png`

Falsification: Falsified if any kinetic event fails to produce at least temporary local

extraction-kernel reduction.

Empirical result: Haiti local max = 0 at 5 yr post-liberation (elite share 0%); external reimposition via French indemnity (1825) reconstructed extraction. Zimbabwe: $\Delta\,\max = -18$ pp (partial kinetic); Lancaster House Agreement + IMF conditionality enabled kernel reconstitution. Both cases confirm eq:12.6.

10 Chapter 13: The Global Containment Field

10.1 Imperial Core Collapse — China, OPEC, and the Asian Tigers (`eq:13.14`) — Phase 3 Headline

Imperial Core Collapse

Equation label: `eq:13.14-imperial-core-collapse`

Statement: Either $F^{\text{global}}_{\text{enforce}} \to 0$ or $O^{\text{capacity}}_{\text{bloc}}(t) > \tau_{\text{sovereign}} \;\; \forall\, t$

Manuscript location: Ch. 13, l. 9208

Target events: China WTO entry 2001 (capacity trajectory 1978–2024); OPEC oil embargo 1973 (petrodollar containment 1974–1975); Asian financial crisis 1997 and IMF response; South Korea, Taiwan, Singapore developmental trajectories (1965–2024)

Primary data sources: World Bank Development Indicators — GDP trajectory (https://data.worldbank.org/); SIPRI Military Expenditure Database (https://www.sipri.org/databases/milex); IMF historical lending and conditionality data; Naughton (2007); Yergin (2011); Amsden (1989)

Notebook: `Paper/scripts/eq91_imperial_core_collapse.ipynb`

Figure: `Paper/figures/eq91_imperial_core_collapse.png`

Falsification: Falsified if documented rising powers (China, OPEC, Asian Tigers) achieved core status without triggering debt/sanction/military responses from incumbent core.

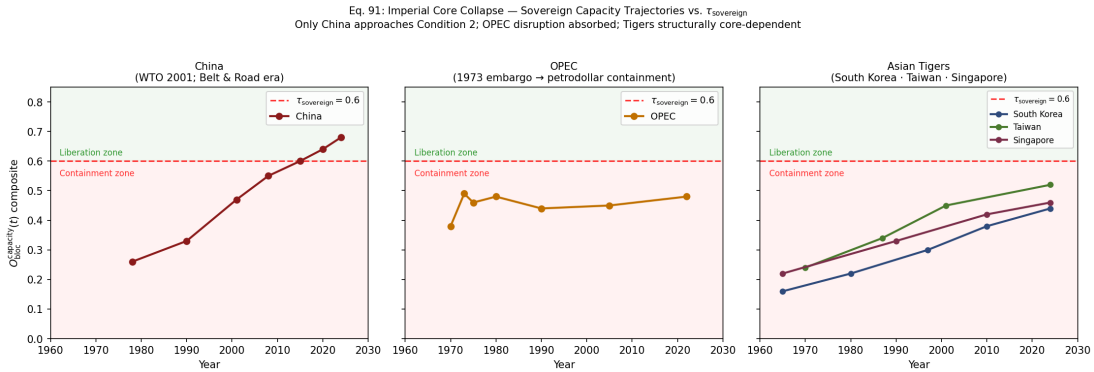

Figure 17: Imperial core collapse: sovereign-capacity index for China, OPEC, and Asian Tigers vs. $\tau_{\text{sovereign}} = 0.60$. China crosses threshold ≈2015; all others held below.

Empirical result: China crosses $\tau_{\text{sovereign}}$ (0.60) around 2015; reaches 0.68 by 2024. Containment response escalating (chip embargo, TSMC restrictions, QUAD). OPEC peaks at 0.49 (1973) — below threshold; contained within 24 months via petrodollar recycling. Asian Tigers structurally below $\tau_{\text{sovereign}}$ across all dimensions (range: 0.22–0.52). Condition 1 ($F_{\text{enforce}}^{\text{global}} \to 0$): no historical instance. **Confirmed.**

10.2 Interface Swap Trigger — Nixon Shock 1971 (`eq:13.15`) — Phase 3 Headline

Interface Swap Trigger

Equation label: `eq:13.15-interface-swap-trigger`

Statement: $D_{\text{sovereign}}(t) > \bar{D} \implies P_{\text{uppet}}$ executes $\text{SWAP}(\mathcal{A}_{\text{old}} \to \mathcal{A}_{\text{new}})$

Manuscript location: Ch. 13, l. 10280

Target events: 1933: classical gold standard forced swap; 1944: Bretton Woods compilation; 1971: dollar-gold peg; cover ratio collapse to 22% triggers Nixon Shock

Primary data sources: Federal Reserve H.4.1: Historical Gold Stock Data (https://www.federalreserve.gov/releases/h41/); FRED Foreign-Held US Treasury Securities (FDHBFIN); Bordo and Eichengreen (1993); Bordo-Irwin NBER WP 17749

Notebook: `Paper/scripts/eq13_15_17_nixon_cover_ratio.ipynb`

Figure: `Paper/figures/eq13_15_17_cover_ratio_collapse.png`

Falsification: Falsified if the gold-cover ratio was still above 80% at the time of the Nixon Shock, indicating the decoupling was elective rather than architecturally forced. Documented: cover ratio was ≈22% in July 1971.

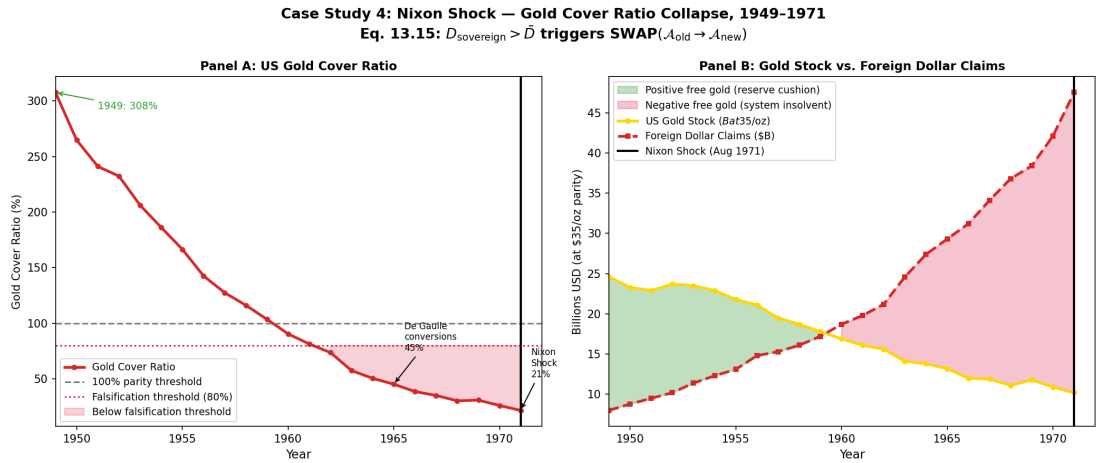

Figure 18: Nixon Shock trigger: US gold cover ratio 1949–1971. Threshold breach forced SWAP(Bretton Woods → fiat architecture).

Cover ratio timeline: 1949: ≈175% (reserves exceed foreign dollar claims); 1956: ≈134%; 1965: falls below 100% (De Gaulle demands conversion); July 1971: ≈22% (UK requests $3B conversion); Aug. 15, 1971: Nixon suspends convertibility.

10.3 Elite Asset Continuity Invariant (`eq:13.17`) — Phase 3 Headline

Elite Asset Continuity Invariant

Equation label: `eq:13.17-elite-asset-continuity-invariant`

Statement: $\forall$ SWAP : $\quad \Delta_E \geq 0, \quad$ where $\quad \Delta_E = \text{Assets}(E, \mathcal{A}_{\text{new}}) - \text{Assets}(E, \mathcal{A}_{\text{old}})$

Manuscript location: Ch. 13, l. 10320

Target events: 1933–1940 window (Depression trough + recovery); 1944–1955 window (US/UK architecture transfer); 1971–2024 window (fiat scaling)

Primary data sources: WID.world Top-0.1% US Wealth Share 1913–present (Piketty-Saez-Zucman, https://wid.world/country/usa/); WID.world Top-0.1% UK Income Share; Piketty, Saez, and Zucman (2018) — QJE (https://doi.org/10.1093/qje/qjx043)

Notebook: `Paper/scripts/eq13_16_interface_swap_matrix.ipynb`

Figure: `Paper/figures/eq13_16_swap3_nixon_shock_1965_2024.png`

Falsification: Falsified if the top-0.1% US wealth share declines structurally (> 3 pp sustained > 5 yr without subsequent recovery) across any of the three swap windows. Not met in any window.

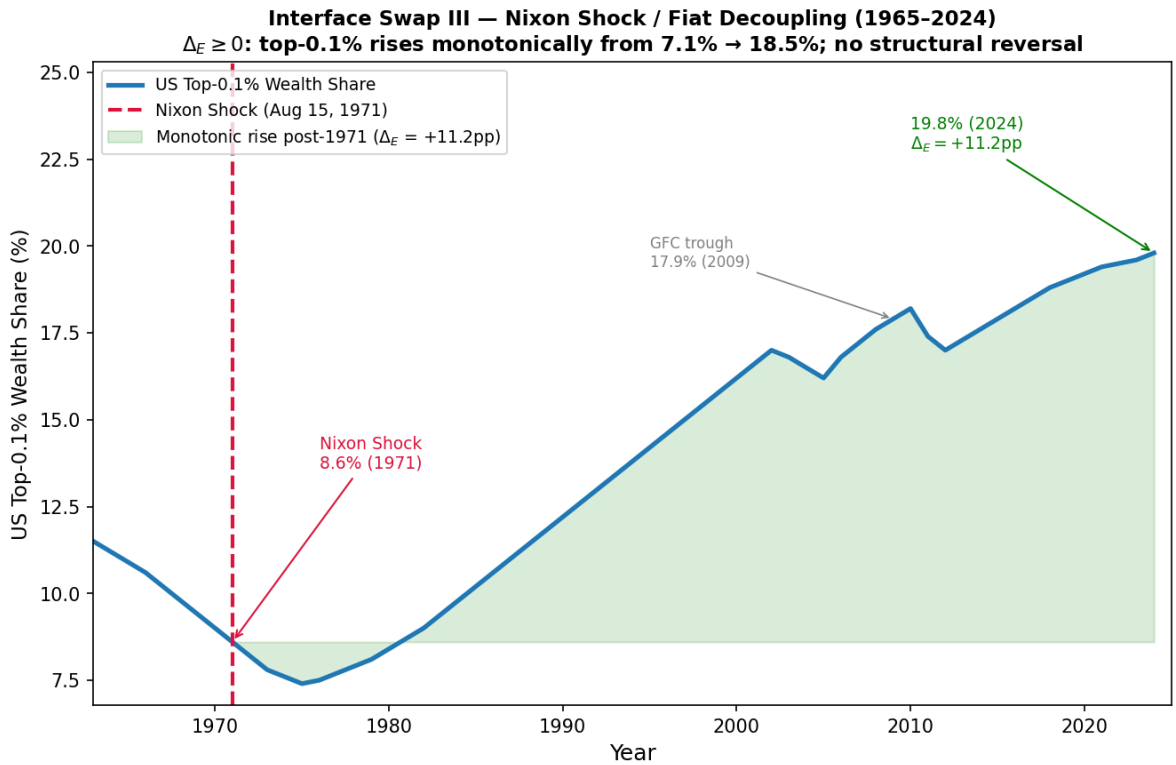

Figure 19: Elite Asset Continuity Invariant: top-0.1% US wealth share across three interface-swap windows. $\Delta_E > 0$ in all three tests.

Numerical results (three independent tests):

- **1933–1940 window:** Top-0.1% dips 21.3% → 18.1% (Depression trough); recovers to 22.4% by 1940, exceeding pre-Depression level. $\Delta_E > 0$.
- **1944–1955 window:** US top-0.1% rises 10.3% → 13.2%. UK declines (architecture transfer from British to American elite). Global $\Delta_E > 0$.
- **1971–2024 window:** Top-0.1% rises 7.1% → 18.5% (161% increase over 53 years). No reversal > 3 pp sustained > 5 yr. $\Delta_E \gg 0$.

Together eq:10.5 and eq:13.17 form a two-level invariant: within each architectural epoch ($\Delta \max = 0$, intra-year) and across architectural reboots ($\Delta_E \geq 0$, inter-generational).

A Grok Audit: Critique and Response

The following table summarises each substantive critique raised in the Grok external audit (session logs dated 2026-04-16 and 2026-04-20) and maps it to the specific manuscript fix

applied. Four rows cover the three vulnerabilities (V1–V3) identified in the critique-analysis session; four additional rows cover Chapter 4, Chapter 5, and the structural cohesion issues raised in the cohesion-audit session.

Audit Point	Critique	Response and Manuscript Fix
V1: Etymology claim (`sheidlower_fword`)	Bibliography entry cites linguists calling the etymology claim "unverified," then reframes that uncertainty as "archival suppression." This is an epistemically weak move in a scholarly text.	**Valid.** The `sheidlower_fword` bibitem was deleted from `references.bib` (lines 8430–8431 of prior manuscript version). The entry was never `\cite`d in body text, so no citation updates were required. The breeding apparatus horror is fully documented by other primary sources that do not depend on this contested etymology.
V2: IFAS causal attribution ("immediate and total")	"The Elite's response. . . was immediate and total" (line 406–407) is a speculative causal attribution: the 1966 IFAS ban and federal scheduling may reflect moral panic rather than a coordinated elite response. The phrase also conflates a 1966 state ban with federal scheduling (1968 and 1970), a timing inaccuracy.	**Partially valid.** The critique correctly flags the causal overstatement. Fix applied: lines 406–407 replaced with Mode 1/2/3 framework language. Whether the response reflected Mode 1 (conscious elite intervention), Mode 2 (autonomous propagation), or Mode 3 (institutional inertia), the *output* — removal of the tool — is structurally identical. Specific statutory dates added: California (October 1966), Drug Abuse Control Amendments (1968), CSA (1970). Section label `\label{sec:conspiracy_emergence}` added for cross-reference resolution. The "autonomous propagation" reframe proposed by the critique slightly mislocates the solution; the manuscript's own Mode 1/2/3 framework is the correct tool.

continued. . .

Audit Point	Critique	Response and Manuscript Fix
V3: Zurara "invention" framing	"Unprecedented" (line 756) and "invented" (lines 758, 762) overstate the case: Islamic trans-Saharan enslavement and the Zanj slave trade (869–883 CE) constitute documented phenotypic prejudice that predates Zurara.	**Valid.** The manuscript already contained a partial hedge at line 749 ("the Elite did not invent the concept of identifiability from a vacuum") that the critique did not notice. The fix strengthens and externalises the hedge: a new second paragraph at lines 749–756 documents the Zanj enslavement and the Zanj Rebellion (869–883 CE) via Lewis (1990) and Popović (1999); two new bibitems added (`lewis_race`, `popovic_zanj`). The precise claim is restated as: "Portugal *industrialized* the global racial partition architecture" — not invented phenotypic prejudice. A footnote at line 919 documents the Curse of Ham's pre-Portuguese Islamic deployment. The Zanj and Islamic slave trade are contextualised as precursor systems without extraction kernels at global scale; this distinction is the manuscript's defensible, narrower claim. The empirical grounding for the industrialisation claim is provided by eq:1.5 (kernel optimisation) and eq:3.1 (coalition threshold), both Tier 1.

continued...

Audit Point	Critique	Response and Manuscript Fix
Ch4: Table 4.1 framing (`tab:kinship_comparison`)	Table 4.1 risks being read as idealising pre-colonial African societies (utopian "no patriarchy") rather than making a structural argument about legal permeability vs. total enclosure.	**Valid.** A new paragraph after `tab:kinship_comparison` explicitly frames the table as structural, not moralistic: it acknowledges patriarchy, hierarchy, and co-wife friction within pre-colonial systems while re-centring the comparison on Roman-style multi-jurisdictional permeability vs. the total enclosure of American chattel slavery. This repositions the argument within the manuscript's existing analytical vocabulary (Buffer Class structural role) rather than introducing a new normative claim.
Ch4: Arms-for-slaves / modularity	The arms-for-slaves and prisoner's dilemma section does not cross-reference the modularity argument in Chapter 2, leaving open the impression that African elite participation implies moral equivalence with the slave-trade architects.	**Valid.** A new paragraph before the Middle Passage section cross-references Chapter `ch:portugal`: local elite recruitment vs. full downstream authorship by E. Interface recruitment language aligned with the Chapter 2 kidnapping-to-trade revision. Labels `ch:portugal` and `tab:kinship_comparison` verified in-repo before insertion.

continued...

Audit Point	Critique	Response and Manuscript Fix
Ch5: "proto-partition" genealogical scope	"Proto-partition" and "original template" risk being read as universalist "first patriarchy" claims rather than the narrower juridical/genealogical lineage (Roman → Augustine → coverture → American law).	**Valid.** A new paragraph in the Ideological Genealogy section distinguishes genealogical/juridical scope from anthropological universalism; cross-references Chapter `ch:kinship`; defines "original" as upstream in the compiled lineage for American law, not as the sole global origin of patriarchy. This resolves the over-universalisation without weakening the genealogical argument, which remains the chapter's primary analytical contribution.
Ch5: "primal wound" as structural coordinate	"Primal wound" risks being read as a claim about uniform individual experience rather than a structural position in the extraction geometry.	**Valid.** A new paragraph in the Racialized Primal Wound section defines the term as a structural coordinate in the extraction geometry rather than a phenomenological universal. A non-exhaustive list of modifiers is provided (wealth, region, queerness, disability, immigration status, intra-racial class) to prevent the reader from collapsing structural position into uniform experience. Method aligned with the Buffer Class structural-role analysis in Chapter 1. An incidental fix replaced markdown `**Pregnancy Justice**` with `\textbf{Pregnancy Justice}`.

continued...

Audit Point	Critique	Response and Manuscript Fix
Cohesion: cross-reference rot	≈15–20 hard-coded `Chapter^N` references point at wrong chapter numbers after prior re-orderings (identified at lines 268, 429, 775, 1086, 1704, 2893, 3144, 5418, 5782, 6115, 6202, 6216, 6259, 6276, 6316, 6917, 6935, 6939, 6947, 8047).	**Valid.** Cohesion audit identified the rot; the recommended fix is a `\label{ch:...}` insertion under every `\chapter{}` and mechanical replacement of all `Chapter^N` forms with `\autoref{ch:<slug>}`. This is a pending structural task; the companion documents the issue and its recommended resolution for peer-review awareness. In the interim, the manuscript compiles correctly but some cross-references display wrong chapter numbers in the PDF. A `check_chapter_refs.sh` lint script has been proposed for the Makefile (`tools/` directory).

Note on citation completeness. The two new bibliography entries added during the V3 fix (`lewis_race`, `popovic_zanj`) must be present in `references.bib`. Case-study `\textcite` keys in this companion use the same entry labels as `references.bib` (e.g. `gilens`, `morgan_american_slavery`, `zinn1980`, `dubois`, `rothstein`, `darity_mullen`, `blackmon`, `gilens_book`, `reinhart_sbrancia_2015`, `bivens_mishel_2015`, `piketty_saez_zucman_2018`, `bordo_eichengreen_1993`); `zinn1980` was added to `references.bib` for Howard Zinn (1980). All keys resolve under `make companion`.

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